

Caixing Wang

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Working Experience

Southeast University

Aug 2025 – Now

Assistant Professor in School of Statistics and Data Science

The Chinese University of Hong Kong

Aug 2024 – July 2025

Postdoc in Statistics

- Mentor: Professor Junhui Wang

Education

Shanghai University of Finance and Economics

Sept 2019 – June 2024

Ph.D. in Statistics

- Supervisor: Professor Xingdong Feng, Associate Professor Xin He

Shanghai University of Finance and Economics

Sept 2015 – June 2019

B.S. in Statistics

Research Interests

Statistical Machine Learning; Kernel Methods; Quantile Regression; Large-scale and Distributed Data Analysis; Deep Learning

Journal Publications (* and † refer to corresponding author and equal contributions (or Alphabet ordering))

- [1]. Estimation of conditional extremiles in reproducing kernel Hilbert spaces with application to large commercial banks data. Fang Chen[†], **Caixing Wang**^{†*}. *Statistica Sinica*, 2025+, accepted.
- [2]. Deep nonparametric quantile regression under covariate shift. Xingdong Feng[†], Xin He^{*†}, Yuling Jiao[†], Lican Kang^{*†}, **Caixing Wang**[†]. *Journal of Machine Learning Research* 25 (385), 1-50.
- [3]. Communication-efficient nonparametric quantile regression via random features. **Caixing Wang**, Tao Li, Xinyi Zhang, Xingdong Feng, Xin He^{*}. *Journal of Computational and Graphical Statistics*, 2024, 33(4), 1175–1184.
- [4]. A lack-of-fit test for quantile regression process models. Xingdong Feng^{*}, Qiaochu Liu, **Caixing Wang**. *Statistics & Probability Letters* 192, 109680, 2023.

Conference Publications

- [5]. Distributed high-dimensional quantile regression: estimation efficiency and support recovery. **Caixing Wang**^{*}, Ziliang Shen. *International Conference on Machine Learning (Spotlight)*, 2024, 235: 51415-51441.
- [6]. Optimal kernel quantile learning with random features. **Caixing Wang**, Xingdong Feng^{*}. *International Conference on Machine Learning (Spotlight)*, 2024, 235: 50419-50452.
- [7]. Towards theoretical understanding of learning large-scale dependent data via random features. Chao Wang, Xin He^{*}, Xin Bing, **Caixing Wang**^{*}. *International Conference on Machine Learning (Spotlight)*, 2024, 235: 50118-50142.
- [8]. Towards a unified analysis of kernel-based methods under covariate shift. Xingdong Feng[†], Xin He[†], **Caixing Wang**^{*†}, Chao Wang[†], Jingnan Zhang[†]. *Advances in Neural Information Processing Systems*, 2023, 36: 73839-73851.

Preprints

- [1]. *Optimal transfer learning for kernel-based nonparametric regression.* Chao Wang[†], **Caixing Wang[†]**, Xin He, Xingdong Feng. **Major Revision in Journal of American Statistical Association.**
- [2]. *Improved statistical analysis for spectral algorithms under general conditions: optimality and robustness.* **Caixing Wang, Qiang Heng, Qi Kuang, Xingdong Feng.** Major Revision in Journal of American Statistical Association.
- [3]. *Generalization properties of robust learning with random features.* **Caixing Wang,** Major Revision in IEEE Transactions on Pattern Analysis and Machine Intelligence.
- [4]. *Estimation and inference on distributed high-dimensional quantile regression: double-smoothing and debiasing.* **Caixing Wang,** Ziliang Shen, Shaoli Wang, Xingdong Feng. **Under review.**
- [5]. *Distributed learning for adaptive and robust nonparametric regression.* **Caixing Wang.** **Under review.**
- [6]. *Inertial quadratic majorization minimization with application to kernel regularized learning.* Qiang Heng, **Caixing Wang***. **Under review.**
- [7]. *High-dimensional differentially private quantile regression: distributed estimation and statistical inference.* Ziliang Shen, **Caixing Wang,** Shaoli Wang, Yibo Yan. **Under review.**
- [8]. *Optimal Sobolev norm learning rates for spectral algorithms with random projections.* **Caixing Wang,** Yue Wang. **Under review.**
- [9]. *Enhancing fairness and efficiency in image classification via weighted SVD-based fine-tuning of large pre-trained models.* **Yixuan Zhang, Jiabin Luo, Caixing Wang,** **Under review.**

Reviewer Services

Journal: *Journal of American Statistical Association (JASA), Journal of the Royal Statistical Society Series B (JRSSB), The Annals of Applied Statistics (AOAS), Journal of Computational and Graphical Statistics (JCGS), Statistica Sinica, Statistics and Its Interface (SII), Econometrics Review (ER), Journal of Parallel and Distributed Computing (JPDC)*

Conference: *International Conference on Learning Representations (ICLR), Neural Information Processing Systems (NeurIPS)*

Open-source Software

- **DisRFBKQR:** R package for “Communication-efficient nonparametric quantile regression via random features” accepted by JCGS.
- **DQR-covariate-shift:** Python package for “Deep nonparametric quantile regression under covariate shift” accepted by JMLR.
- **DHSQR:** R package for “Distributed high-dimensional quantile regression: estimation efficiency and support recovery” accepted by ICML.
- **Kernel-CS:** Python package for “Towards a unified analysis of kernel-based methods under covariate shift” accepted by NeurIPS.

Talks

International Conference on Statistics, Data Science and Artificial Intelligence (2024) *CCUT, Changchun*

- Deep quantile regression under covariate shift

The 1st International Conference for PhD Pioneers (2024) *RUC, Beijing*

- Optimal transfer learning for kernel-based nonparametric regression

The 2nd Conference for Chinese Statistical Association of Young Scholars (2024) *JNU, Xuzhou*

- Towards a unified analysis of kernel-based methods under covariate shift

Teaching Experience

Teaching Assistant *2020-2024*

- Undergraduate Courses: Survival analysis, Extreme value theory.

Skills

- **Language:** Strong reading, writing and speaking competencies for English and Chinese.
- **Programming:** Python, R, C++.
- **Operating Systems:** Windows, MacOS, Linux.
- **Documentation:** LATEX, Markdown, MS Office.